

Crossing truck

Moving in the world frame: **the white box truck**. Staying in place: 2 yellow tactile pavings, the traffic light pole, 2 green railings.

The white box truck moves about 16.4 m in the world frame at an average speed of 6.31 m/s; it accelerates over the clip; it reverses direction around frame 19.

Question

Right answer: D

In the world frame (ignoring camera motion), approximately how far (in meters) did **the white box truck** move between frame 7 and frame 14 of 34?

A. 12 B. 6.4 C. 4.7 **D. 8.7**

ground truth: 8.14 m

Qwen3-VL-8B · frames only

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<evidence>
object: the white box truck
position_frame_7_m: (4.1, 1.0, 9.3)
position_frame_14_m: (-0.2, -0.2, 10.0)
displacement_m: 4.5
</evidence>
<answer>C</answer>
```

Ours · frames + trajectory tokens

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<evidence>
object: the white box truck
position_frame_7_m: (5.6, 2.6, 13.6)
position_frame_14_m: (-2.7, 2.3, 14.8)
displacement_m: 8.4
</evidence>
<answer>D</answer>
```